

ASEO: A Method for the Simultaneous Estimation of Single-Trial Event-Related Potentials and Ongoing Brain Activities

Luzhou Xu, *Member, IEEE*, Petre Stoica, *Fellow, IEEE*, Jian Li, *Fellow, IEEE*, Steven L. Bressler, Xianzhi Shao, and Mingzhou Ding*, *Member, IEEE*

Abstract—Cognitive functions are often studied by recording electric potentials from the brain over repeated presentations of a sensory stimulus or repeated performance of a motor action. Each repetition is called a trial. Recent work has demonstrated that contrary to the traditional view, the event-related potential (ERP) can vary from trial to trial and the background ongoing activity often contains rich information about the cognitive state of the brain. Based on such a variable signal plus ongoing activity model, an iterative parameter estimation method is proposed in which both the single-trial parameters of the ERP and the autoregressive representation of the ongoing activity are obtained simultaneously. This technique, referred to as the analysis of single-trial ERP and ongoing activities method, is first tested on simulation examples, and then applied to the local field potential recordings from monkeys performing a visuomotor task.

Index Terms—Autoregressive (AR), electroencephalogram (EEG), event-related potential (ERP), local field potential (LFP), maximum-likelihood estimate (MLE), ongoing brain activity.

I. INTRODUCTION

IN A TYPICAL cognitive neuroscience experiment, scalp-recorded electroencephalograms (EEGs) or invasively recorded local field potentials (LFPs) are traditionally modeled as a linear combination of a stereotypic signal that is time-locked to the stimulus onset (movement onset) and a noise background. According to this signal-plus-noise (SPN) model, the signal is extracted by performing an average across an ensemble of trials to eliminate the noise [1], [2]. The recovered signal is called the event-related potential (ERP) and is a powerful

index of cognitive brain functions [3], [4]. Recent work has carefully examined the two assumptions underlying the SPN model and found them to be overly simplistic [5]. First, the ERP can vary from trial to trial even when the experimental conditions are kept the same. Second, the background is not noise despite its noise-like appearance but represents ongoing brain activity rich in oscillatory content, providing another index of cognitive brain functions [6]. These new insights have led to a new class of models of event-related brain recordings that we call the variable signal plus ongoing activity (VSPOA) models [7].

Over the years, VSPOA models with varying degrees of complexity have appeared in the literature [8]–[10]. Most models treat the entire evoked potential as one unitary waveform whose amplitude and latency vary from trial to trial. Truccolo *et al.* [11] proposed a more realistic VSPOA model in which the LFP from a recording site is modeled as a linear combination of multiple ERP components as well as ongoing brain activity. In this model, each ERP component is assumed to have a deterministic but unknown characteristic waveform. The amplitudes and latencies of different ERP components vary from trial to trial as well as with respect to one another. As shown in [11], better estimates of the ERPs can be obtained based on this model. More importantly, this model allows the investigation of the single-trial latency and amplitude variations of each component, which are themselves physiological variables containing information about event-related information processing in the brain. A multichannel extension of the Truccolo *et al.* model has recently appeared in [12].

A number of techniques have been developed to estimate the ERP on a single-trial basis, including the matched-filter method [8], [13], the maximum-likelihood estimator (MLE) [14]–[16], the wavelet transform approach [17]–[21], and clustering analysis [22]. In particular, the differentially variable component analysis (dVCA) method proposed in [11] is based on the VSPOA model with multiple ERP components. This approach estimates each ERP component's waveform, and its single-trial latency shift and amplitude scaling factor by using the components' differential variability from trial to trial. In the derivation of the dVCA method [11], the ongoing activity is treated as a white Gaussian noise with zero mean and an (unknown) time-independent variance. As indicated earlier, this white noise assumption, in general, is not true and may affect the performance of the estimation algorithm. Moreover, the dVCA method is implemented in the time domain. Due to the matched-filtering

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L. Xu is with ArrayComm LLC, San Jose, CA 95131 USA.

P. Stoica is with the Department of Information Technology, Uppsala University, Uppsala 75105, Sweden.

J. Li is with the Department of Electrical and Computer Engineering, University of Florida, Gainesville, FL 32611 USA.

S. L. Bressler is with the Center for Complex Systems, Florida Atlantic University, Boca Raton, FL 33431 USA.

X. Shao was with J. Crayton Pruitt Family Department of Biomedical Engineering, University of Florida, Gainesville, FL 32611 USA. He is now with OnCure Medical Corporation, Newport Beach, CA 92660 USA.

*M. Ding is with J. Crayton Pruitt Family Department of Biomedical Engineering, University of Florida, Gainesville, FL 32611 USA (e-mail: mding@bme.ufl.edu).

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operation in the method, the resolution of the ERP latency estimates is constrained to be integer multiplies of the sampling interval, which could be considered another weakness of the method.

The problem of estimation inaccuracy due to discrete temporal sampling can be overcome if the estimation is formulated in the frequency domain. Two frequency-domain methods have been presented in [9] and [10]. In these methods, the single-trial latency shifts are no longer required to be estimated as integer multiplies of the sampling period, and hence, the estimation accuracy can be improved. Furthermore, it is known that under a mild condition [23], the Fourier transforms (FTs) of the ongoing activity for different frequencies can be considered as uncorrelated. This statistical property has been exploited in [9] and [10], where the ongoing activities at different frequencies are modeled as independent zero-mean Gaussian random variables whose variances [power spectral density (PSD)] are treated as unknowns to be estimated. Based on this idea, two similar iterative MLEs are developed by using the cyclic maximization technique [24]. However, it can be shown that the formulations of the estimation problems in [9] and [10] are not really appropriate since the corresponding MLEs do not, in fact, exist [25], [26]. Specifically, the likelihood functions associated with the estimation problems in [9] and [10] have no maxima, but only saddle points.

In this paper, we propose an iterative method, based on the VSPOA model proposed in [11], which estimates the single-trial parameters of the ERP components and the ongoing activity simultaneously. In contrast to the methods in [9] and [10], the ongoing activity is herein assumed to be an autoregressive (AR) random process. This assumption reduces the number of unknowns significantly, and the resulting MLE avoids the nonexistence problem mentioned before. Our method has two steps. First, the waveforms of the ERP components and their single-trial latency shifts and the amplitude scaling factors are estimated in the frequency domain. Second, the parameters of the ongoing AR random process are determined in the time domain from the residuals obtained by removing the estimated single-trial ERPs from the single-trial LFPs. These two steps are iterated until no further improvement is seen in the estimated quantities. We refer to this method as the analysis of single-trial ERP and ongoing activity (ASEO) method. We test the performance of the ASEO method using simulated examples, and then apply it to an experimental dataset consisting of intracortically recorded LFPs in a monkey performing a GO/NOGO visuomotor pattern discrimination task.

The remainder of this paper is organized as follows. In Section II, we describe the VSPOA model. The ASEO method is then formulated in Section III. To facilitate the comparison between the ASEO method and the dVCA method, we provide a brief summary of the dVCA method in Section IV. In Section V, we demonstrate the effectiveness of the ASEO method by applying it to several simulated and experimental examples. Finally, Section VI contains our conclusions.

II. DATA MODEL AND PROBLEM FORMULATION

To fix the terminology, our theoretical development next is formulated in terms of the LFP. Similar development can be carried out for the EEG. Let the single-trial LFP be described by the following VSPOA model:

$$x_r(t) = \sum_{n=1}^N \beta_{rn} s_n(t - \tau_{rn}) + z_r(t) \quad (1)$$

where $x_r(t)$ ($r = 1, 2, \dots, R$ and $t = 0, 1, \dots, T-1$) is the observed LFP discrete-time signal on the r th trial (R and T being the number of trials and the number of data samples for each trial, respectively), $s_n(t)$ ($n = 1, 2, \dots, N$) is the unknown ERP component waveform (N being the total number of components), β_{rn} and τ_{rn} are the unknown single-trial amplitude scaling factor and latency phase shift, respectively, and $z_r(t)$ is the ongoing activity. In this paper, we assume that the number of the ERP components N is given. In applications, N can be chosen by inspecting the average ERP (AERP) waveform, or it can be determined by using an information theoretic criterion [27]. The ongoing activity $z_r(t)$ in (1) is assumed to be an AR random process

$$z_r(t) = \sum_{k=1}^K a_k z_r(t-k) + e_r(t) \quad (2)$$

where a_k ($k = 1, 2, \dots, K$) is the k th unknown AR coefficient, K is the AR model order, and $e_r(t)$ is a white Gaussian noise with zero mean and unknown variance σ^2 . The problem of interest is to estimate the unknown ERP component waveforms $\{s_n(t)\}$, the corresponding single-trial amplitude scaling factors $\{\beta_{rn}\}$ and latency shifts $\{\tau_{rn}\}$, and the AR coefficients $\{a_k\}$ of the ongoing activity $z_r(t)$.

By applying the discrete FT (DFT) to (1), we obtain the frequency-domain representation of (1) as follows:

$$X_r(\omega) = \sum_{n=1}^N \beta_{rn} e^{-j\omega\tau_{rn}} S_n(\omega) + Z_r(\omega) \quad (3)$$

where $X_r(\omega)$, $S_n(\omega)$, and $Z_r(\omega)$ stand for the FTs of $x_r(t)$, $s_n(t)$, and $z_r(t)$, and $\omega \in \{0, \omega_0, 2\omega_0, \dots, (T-1)\omega_0\}$ is the radian frequency with $\omega_0 = 2\pi/T$. Note that $x_r(t)$, $s_n(t)$, and $z_r(t)$ in (1) are all real-valued functions of t , and hence, their FTs are Hermitian functions of ω . In other words, we have $X_r(\omega) = X_r^*(2\pi - \omega)$, $S_n(\omega) = S_n^*(2\pi - \omega)$, and $Z_r(\omega) = Z_r^*(2\pi - \omega)$ with $(\cdot)^*$ denoting the complex conjugate. Therefore, we consider only the frequency samples at $\omega \in \{0, \omega_0, 2\omega_0, \dots, \lceil T/2 \rceil \omega_0\}$ with $\lceil T/2 \rceil$ denoting the smallest integer larger than $T/2$. From (2), we know that $Z_r(\omega)$ for different values of r and ω are independently and identically distributed (i.i.d.) circularly symmetric complex Gaussian random variables with zero mean and unknown variance $f(\omega)$, where

$$f(\omega) = \frac{\sigma^2}{\left| 1 - \sum_{k=1}^K a_k e^{-jk\omega} \right|^2} \quad (4)$$

is the PSD of the ongoing activity in (2) (see, e.g., [23]).

III. ASEO ALGORITHM

Our parameter estimation problem is formulated in terms of the VSPOA model in (1). This model can be seen as a direct extension of the classical SPN model $x_r(t) = s(t) + z_r(t)$ by noting that: 1) $s(t)$ is not invariant from trial to trial and 2) it may contain several components (e.g., sensory versus motor) that vary with respect to one another on a trial by trial basis. It is important to recognize that this model is a phenomenological model and it does not address the origin of the ERPs. The genesis of ERPs is a vigorously debated issue [28]–[30]. Whereas one school of thought maintains that the evoked potential is the result of the prestimulus ongoing activity being phase-reset by the stimulus [31]–[36], others present evidence arguing against this proposition [37], [38]. The VSPOA model in (1) does not depend critically on the outcome of this debate. It mainly assumes that there are activities in the data that are relatively phase-locked to the stimulus onset or movement onset, irrespective of whether this phase locking comes from phase resetting or the firing of new cell groups. However, regarding the ongoing activity term, the phase resetting hypothesis implies that $z_r(t)$ are different processes during pre- and poststimulus time periods, namely, it is nonstationary. Examples are used in Section V to demonstrate that ASEO is still effective under such circumstances.

From the frequency-domain data model in (3), we know that $\{X_r(\omega)\}_{\omega,r}$ are i.i.d. circularly symmetric complex Gaussian random variable with mean $\sum_{n=1}^N \beta_{rn} e^{-j\omega\tau_{rn}} S_n(\omega)$ and variance $f(\omega)$. The joint probability density function (PDF) of $\{X_r(\omega)\}_{\omega,r}$ can be written as

$$\prod_{r=1}^R \prod_{\omega=0}^{\lceil T/2 \rceil \omega_0} \frac{1}{2\pi f(\omega)} \times \exp \left\{ -\frac{|X_r(\omega) - \sum_{n=1}^N \beta_{rn} e^{-j\omega\tau_{rn}} S_n(\omega)|^2}{f(\omega)} \right\}. \quad (5)$$

Applying logarithm to (5) and multiplying the result by a negative sign yield the negative log-likelihood function of the LFP data (to within a additive constant) as follows:

$$L = \sum_{r=1}^R \sum_{\omega=0}^{\lceil T/2 \rceil \omega_0} \{ \ln f(\omega) + f^{-1}(\omega) \left| X_r(\omega) - \sum_{n=1}^N \beta_{rn} e^{-j\omega\tau_{rn}} S_n(\omega) \right|^2 \} \quad (6)$$

where $f(\omega)$ is defined as a function of $\{a_k\}$ and σ^2 in (4). Minimizing (6) with respect to $\{S_n(\omega)\}$, $\{\beta_{rn}\}$, $\{\tau_{rn}\}$, $\{a_k\}$, and σ^2 yields the MLEs of the unknown parameters. However, in general, the minimizer of (6) cannot be produced in closed form. Consequently, we propose an iterative method, namely, the ASEO method referred to earlier to solve this problem. The proposed ASEO algorithm includes two main steps: the F (frequency) step and the T (time) step, which are iterated. In the F-step, using the most recently available estimate of the PSD of the ongoing activity, we estimate (in the frequency domain) the ERP parameters by minimizing the negative log-likelihood

function in (6) with respect to $\{S_n(\omega)\}$, $\{\beta_{rn}\}$, and $\{\tau_{rn}\}$. In the T-step, we estimate (in the time domain) the parameters of the ongoing activity based on the most recently estimated ERP parameters. Note that unlike [9] and [10], the PSD of the ongoing activity in (6) is (for identifiability reasons) constrained to correspond to an AR process, which makes it difficult to estimate $f(\omega)$ directly in the frequency domain. Instead, we consider the inverse DFTs of the residual LFPs, which are obtained by removing the estimated ERP components from LFPs. By using available methods [23], the ongoing activity model parameters, including the AR coefficients $\{a_k\}$, and the white noise variance σ^2 can be estimated readily from the residual signals in the time domain. The final estimates of the parameters of the ERPs and the ongoing activity are obtained by iterating the aforementioned two steps until convergence.

We give shortly a detailed description of the ASEO algorithm.

A. F-Step

In the F-step, we estimate the ERP parameters based on an initial or intermediate estimate of the PSD of the ongoing activity. This includes three substeps for estimating the ERP component waveforms, the single-trial latency shifts, and the single-trial amplitude scaling factors.

1) *Estimation of the ERP Component Waveforms*: In this substep, we estimate the ERP component waveforms $\{S_n(\omega)\}$ based on initial or intermediate estimates of the ERP single-trial amplitude scaling factors and latency shifts.

Let

$$\mathbf{X}(\omega) = [X_1(\omega) \ X_2(\omega) \ \cdots \ X_R(\omega)]^T \quad (7)$$

$$\mathbf{S}(\omega) = [S_1(\omega) \ S_2(\omega) \ \cdots \ S_N(\omega)]^T \quad (8)$$

$$\hat{\mathbf{H}}(\omega) = [\hat{\mathbf{h}}_1(\omega) \ \hat{\mathbf{h}}_2(\omega) \ \cdots \ \hat{\mathbf{h}}_N(\omega)] \quad (9)$$

where

$$\hat{\mathbf{h}}_n(\omega) = [\hat{\beta}_{1n} e^{-j\hat{\tau}_{1n}\omega} \ \hat{\beta}_{2n} e^{-j\hat{\tau}_{2n}\omega} \ \cdots \ \hat{\beta}_{Rn} e^{-j\hat{\tau}_{Rn}\omega}]^T. \quad (10)$$

Hereafter, $(\cdot)^T$ denotes the transpose, whereas $\{\hat{\beta}_{rn}\}$ and $\{\hat{\tau}_{rn}\}$ are the initial or intermediate estimates of $\{\beta_{rn}\}$ and $\{\tau_{rn}\}$, respectively.

From (6), the ML estimate of $\{S_n(\omega)\}$, given $\{\hat{\beta}_{rn}\}$ and $\{\hat{\tau}_{rn}\}$, can be obtained as follows:

$$\hat{\mathbf{S}}(\omega) = \arg \min_{\mathbf{S}(\omega)} \|\mathbf{X}(\omega) - \hat{\mathbf{H}}(\omega)\mathbf{S}(\omega)\|^2 \quad (11)$$

with $\|\cdot\|$ being the Euclidean norm. Solving the optimization problem in (11) yields

$$\hat{\mathbf{S}}(\omega) = [\hat{\mathbf{H}}^H(\omega)\hat{\mathbf{H}}(\omega)]^{-1}\hat{\mathbf{H}}^H(\omega)\mathbf{X}(\omega) \quad (12)$$

with $(\cdot)^H$ denoting the conjugate transpose. Computing (12) for each ω of interest, i.e., $\omega = 0, \omega_0, 2\omega_0, \dots, \lceil T/2 \rceil \omega_0$, yields the (intermediate) estimates of the ERP component waveforms in the frequency domain.

2) *Estimation of the Single-Trial Latency Shifts*: Since closed-form estimates of $\{\tau_{rj}\}$ cannot be produced, in this substep, we estimate $\{\tau_{rj}\}$ one by one using the maximization (CM) technique [24]. Within each step of the CM algorithm, we

estimate one latency shift τ_{rn} by using the initial or intermediate estimates of the others, i.e., $\{\tau_{rj}\}_{j \neq n}$.

Let

$$\tilde{X}_{r,n}(\omega) = X_r(\omega) - \sum_{j \neq n} \hat{\beta}_{rj} e^{-j\omega \hat{\tau}_{rj}} \hat{S}_j(\omega) \quad (13)$$

where $\{\beta_{rj}\}$ and $\{\tau_{rj}\}_{j \neq n}$ are the most recent estimates of the single-trial amplitudes and the single-trial latency shifts. From (6), the estimate of τ_{rn} can be obtained by solving the following optimization problem:

$$\hat{\tau}_{rn} = \arg \min_{\tau_{rn}} \sum_{\omega=0}^{\lceil T/2 \rceil \omega_0} \left\{ \hat{f}^{-1}(\omega) \times \left| \tilde{X}_{r,n}(\omega) - e^{-j\omega \tau_{rn}} \hat{\beta}_{rn} \hat{S}_n(\omega) \right|^2 \right\} \quad (14)$$

where $f(\omega)$ is the most recent estimate of the power spectrum of the ongoing activities. After some straightforward manipulations, (14) can be simplified as

$$\hat{\tau}_{rn} = \arg \max_{\tau_{rn}} \operatorname{Re} \left\{ \sum_{\omega=0}^{\lceil T/2 \rceil \omega_0} e^{j\omega \tau_{rn}} \tilde{X}_{r,n}(\omega) \hat{f}^{-1}(\omega) \hat{S}_n^*(\omega) \right\} \quad (15)$$

where $(\cdot)^*$ and $\operatorname{Re}(\cdot)$ denote the complex conjugate and the real part, respectively. Note that the cost function in (15) is the real part of the inverse DFT of

$$\tilde{X}_{r,n}(\omega) \hat{f}^{-1}(\omega) \hat{S}_n^*(\omega)$$

and can be calculated by using inverse fast FT (IFFT) efficiently.

The estimate of the single-trial latency shifts can be summarized as follows:

Do for $n = 1, 2, \dots, N$ the following steps.

- 1) *Compute $\tilde{X}_{r,n}(\omega) \hat{f}^{-1}(\omega) \hat{S}_n^*(\omega)$ for each ω .*
- 2) *Apply IFFT to $\tilde{X}_{r,n}(\omega) \hat{f}^{-1}(\omega) \hat{S}_n^*(\omega)$.*
- 3) *Search for the maximum of the real part of the IFFT output.*

Note that the aforementioned algorithm can be iterated until convergence to obtain the exact optimal $\{\tau_{rj}\}$ [given $\{\beta_{rj}\}$ and $f(\omega)$]. However, since this algorithm is embedded in an iterative algorithm, exact optimal $\{\tau_{rj}\}$ are not necessary. Hence, we do not need to iterate the earlier procedure.

3) *Estimation of the Single-Trial Amplitude Scaling Factors:*

In this substep, we let

$$\mathbf{X}_r = \left[X_r(0), X_r(\omega_0), \dots, X_r\left(\left\lceil \frac{T}{2} \right\rceil \omega_0\right) \right]^T \quad (16)$$

$$\boldsymbol{\beta}_r = [\beta_{r1} \ \beta_{r2} \ \dots \ \beta_{rN}]^T \quad (17)$$

$$\hat{\mathbf{G}}_r = [\hat{\mathbf{g}}_{r1} \ \hat{\mathbf{g}}_{r2} \ \dots \ \hat{\mathbf{g}}_{rN}] \quad (18)$$

$$\hat{\mathbf{g}}_{rn} = \left[\hat{S}_n(0) \ \hat{S}_n(\omega_0) e^{-j\hat{\tau}_{rn}\omega_0} \ \dots \ \hat{S}_n(\lceil T/2 \rceil \omega_0) e^{-j\hat{\tau}_{rn} \lceil T/2 \rceil \omega_0} \right]^T \quad (19)$$

and

$$\hat{\mathbf{F}} = \operatorname{diag} \left(\hat{f}(0), \hat{f}(\omega_0), \dots, \hat{f}(\lceil T/2 \rceil \omega_0) \right) \quad (20)$$

where $\operatorname{diag}(\cdot)$ denotes a diagonal matrix.

By using (6) and the fact that $\{\beta_{rn}\}$ are real numbers, the estimates of $\{\beta_{rn}\}$ can be obtained as

$$\begin{aligned} \hat{\beta}_r &= \arg \min_{\beta_r} (\mathbf{X}_r - \hat{\mathbf{G}}_r \boldsymbol{\beta}_r)^H \hat{\mathbf{F}}^{-1} (\mathbf{X}_r - \hat{\mathbf{G}}_r \boldsymbol{\beta}_r) \\ &\text{subject to } \boldsymbol{\beta}_r \in \mathbb{R}^N. \end{aligned} \quad (21)$$

Solving the optimization problem in (11) gives

$$\hat{\beta}_r = [\operatorname{Re}(\hat{\mathbf{G}}_r^H \hat{\mathbf{F}}^{-1} \hat{\mathbf{G}}_r)]^{-1} \operatorname{Re}(\hat{\mathbf{G}}_r^H \hat{\mathbf{F}}^{-1} \mathbf{X}_r). \quad (22)$$

B. T-Step

In this step, we consider the identification of the ongoing activity, given initial or intermediate estimates of the ERP parameters. Note that to avoid the problem of a nonexistent MLE, which occurs in the formulations in [9] and [10], in this paper, the ongoing activity is modeled as an AR random process, i.e., the PSD $f(\omega)$ of the ongoing activity is constrained by (4). The AR parameters can be estimated in the frequency domain directly by inserting (4) into (6) and minimizing (6) with respect to $\{a_k\}$ and σ^2 . However, the frequency-domain method is computationally intensive. Instead, we consider next estimating the AR parameters in the time domain using an approximate ML method.

Let $\hat{Z}_r(\omega)$ be the FT of the residual LFP, i.e.

$$\hat{Z}_r(\omega) = X_r(\omega) - \sum_{n=1}^N \hat{\beta}_{rn} e^{-j\hat{\tau}_{rn}\omega} \hat{S}_n(\omega). \quad (23)$$

Let $\hat{z}_r(t)$ be the inverse DFT transform of $\hat{Z}_r(\omega)$. Note that $\hat{z}_r(t)$ is real valued due to the Hermitian symmetry of $\hat{Z}_r(\omega)$. Then, $\hat{z}_r(t)$ is approximately an AR process, i.e.

$$\hat{z}_r(t) = \sum_{k=1}^K a_k \hat{z}_r(t-k) + e_r(t). \quad (24)$$

We assume that the AR order K is given. In practice, the value of K can be chosen based on physiological considerations or determined by using, for example, Bayesian information criteria (BIC). To estimate the AR coefficients $\{a_k\}$ and σ^2 , let

$$\mathbf{a} = [a_1 \ a_2 \ \dots \ a_K]^T \quad (25)$$

$$\mathbf{y}_r = [\hat{z}_r(K) \ \hat{z}_r(K+1) \ \dots \ \hat{z}_r(T-1)]^T \quad (26)$$

and

$$\mathbf{Y}_r = \begin{bmatrix} \hat{z}_r(K-1) & \hat{z}_r(K-2) & \dots & \hat{z}_r(0) \\ \hat{z}_r(K) & \hat{z}_r(K-1) & \dots & \hat{z}_r(1) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{z}_r(T-2) & \hat{z}_r(T-3) & \dots & \hat{z}_r(T-1-K) \end{bmatrix}. \quad (27)$$

From (24), $\mathbf{y}_r$ can be written in the form

$$\mathbf{y}_r = \mathbf{Y}_r \mathbf{a} + \mathbf{e}_r \quad \text{with} \quad r = 1, 2, \dots, R \quad (28)$$

where $\mathbf{e}_r$ is defined similarly to $\mathbf{y}_r$. From (28), we obtain estimates of $\mathbf{a}$ and σ^2 via the ML method

$$\hat{\mathbf{a}} = \left(\sum_{r=1}^R \mathbf{Y}_r^T \mathbf{Y}_r \right)^{-1} \left(\sum_{r=1}^R \mathbf{Y}_r^T \mathbf{y}_r \right) \quad (29)$$

and

$$\hat{\sigma}^2 = \frac{1}{R(T-K)} \sum_{r=1}^R \|\mathbf{y}_r - \mathbf{Y}_r \hat{\mathbf{a}}\|^2. \quad (30)$$

Once we have obtained the estimates of $\{a_k\}$ and σ^2 , the PSD estimate of the ongoing activity can be calculated readily by using (4).

C. Initialization and Ambiguity Elimination

Before iterating the F-step and T-step, we need to choose the initial estimates. In this regard, features of the average AERP can serve as a guide. (Here, AERP is obtained by performing a traditional ensemble average triggered on stimulus or movement onset.) Specifically, following [11], we determine the number of components N based on the inspection of the AERP, and then, choose N segments of the AERP as the initial estimates of the ERP component waveforms. The DFTs of the so-obtained segments of the AERP are used as the initial estimates of $\{S_n(\omega)\}$. In some cases, enhanced initial estimates of the ERP components can be obtained based on some physiological considerations. The initial estimates for the single-trial amplitude scaling factors and latency shifts are set to 1 and 0, respectively. The ongoing activity is initialized as a temporally white Gaussian noise, i.e., $\hat{f}(\omega) = 1$.

We note that the representation of LFPs in (1) or (3) is not unique. For example, the LFPs in the frequency domain in (3) can be rewritten as

$$X_r(\omega) = \sum_{n=1}^N \frac{\beta_{rn}}{\Delta_n^\beta} e^{-j\omega(\tau_{rn} + \Delta_n^\tau)} [\Delta_n^\beta e^{j\omega \Delta_n^\tau} S_n(\omega)] + Z_{rn}(\omega). \quad (31)$$

In (31), the ERP component waveforms $\{\Delta_n^\beta e^{j\omega \Delta_n^\tau} S_n(\omega)\}$, the corresponding single-trial amplitude scaling factors $\{\beta_{rn}/\Delta_n^\beta\}$, and the latency shifts $\{\tau_{rn} + \Delta_n^\tau\}$ are ambiguous due to the arbitrariness of Δ_n^β and Δ_n^τ . To eliminate these ambiguities, we constrain the maximum magnitude of the ERP components to be 1 and the mean of the single-trial latency shifts to be 0

$$\max_t |\hat{s}_n(t)| = 1 \quad \text{and} \quad \sum_{r=1}^R \hat{\tau}_{rn} = 0, \quad \text{for } n = 1, 2, \dots, N \quad (32)$$

where $\hat{s}_n(t)$ is the inverse DFT of $\hat{S}_n(\omega)$. The final estimates of the ERP parameters can be adjusted to satisfy these two constraints.

A step-by-step summary of the ASEO method is as follows.

1) Initialization:

- a) Choose N AERP segments as the initial estimates of the ERP component waveforms (see [11]). Calculate the DFTs of the so-obtained AERP segments

and the recorded LFPs. Set $\hat{\beta}_{rn} = 1$, $\hat{\tau}_{rn} = 0$, and $\hat{f}(\omega) = 1$.

- b) Update the estimates of the single-trial latency shifts and amplitude scaling factors using the CM algorithm in Section III-A2 and (22), respectively.
- 2) F-step:
 - a) Update the estimates of the ERP component waveforms for $\omega = 0, \omega_0, 2\omega_0, \dots, \lceil T/2 \rceil \omega_0$ by using (12).
 - b) Update the estimates of the single-trial latency shifts using the CM algorithm in Section III-A2 for each trial and each component.
 - c) Update the estimates of the single-trial amplitude scaling factors for each trial using (22).
- 3) T-step:
 - a) Calculate the frequency-domain residual LFPs $\hat{Z}_r(\omega)$ by removing the estimated ERP components from $X_r(\omega)$, and then, calculate their inverse DFTs to get $\hat{z}_r(t)$.
 - b) Estimate $\{a_k\}$ and σ^2 using (29) and (30).
 - c) Calculate the PSD of the ongoing activity using (4).
- 4) Iteration: Iterate the F-step and T-step before a number of times.
- 5) Ambiguity elimination: Calculate the time-domain ERP component waveforms using the inverse DFT, and then adjust the ERP component waveforms, single-trial amplitude scaling factors, and latency shifts to satisfy the constraints in (32).

IV. DVCA ALGORITHM

We claim that the ASEO method represents a significant improvement, both theoretically and in practical applications, over the previously published differentially dVCA algorithm [11]. Comparison will be made in the next section to demonstrate this point. Basically, dVCA is a least-square (LS) estimator. It can be formulated as follows:

$$\min_{\{\beta_r, n\}, \{\tau_{rn}\}, \{s_n(t)\}} \sum_{t=1}^T \sum_{r=1}^R \left| x_r(t) - \sum_{n=1}^N \beta_{rn} s_n(t - \tau_{rn}) \right|^2. \quad (33)$$

Since the aforementioned optimization problem does not yield a closed-form solution, the authors in [11] developed an iterative algorithm by using the cyclic maximization technique [24]. The dVCA algorithm consists of the following steps.

- 1) Choose N AERP segments as the initial estimates of the ERP component waveforms.
- 2) For all trials and components, estimate the single-trial latency shifts.
- 3) Estimate the ERP waveforms component by component.
- 4) For all trials and components, estimate the single-trial amplitudes.
- 5) Go to step 2 and repeat until convergence.

Note that the whole dVCA algorithm is implemented in the time domain. Also, note that the dVCA algorithm does not employ the ongoing activity model in (2). Actually, as an LS method, it implicitly assume that the ongoing activities are

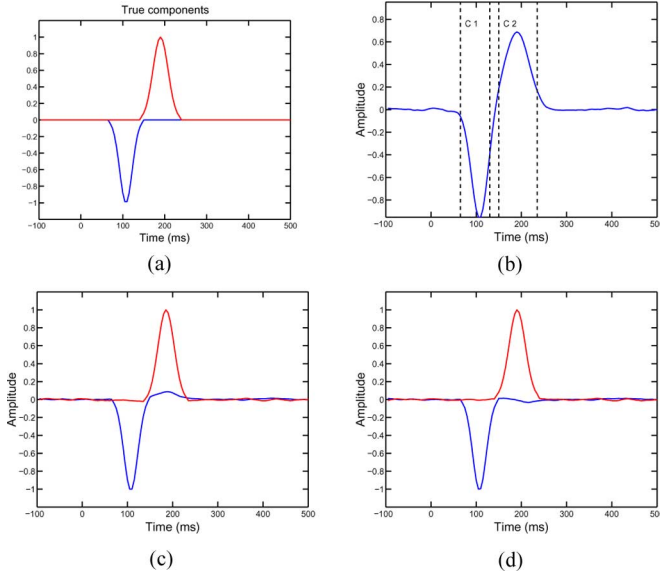


Fig. 1. ERP and AERP waveforms when there are two ERP components. (a) True ERP component waveforms. (b) AERP. (c) ERP component waveforms estimated by dVCA. (d) ERP component waveforms estimated by ASEO.

independently and i.i.d. white Gaussian noise, which, in general, is not true and may adversely affect the performance of the estimation algorithm.

V. NUMERICAL AND EXPERIMENTAL EXAMPLES

We shortly present simulated and experimental examples to demonstrate the performance of the proposed ASEO algorithm.

A. Simulated Data

1) *Example 1:* In this example, we assume that $R = 220$ trials of LFP signals, each with $T = 120$ data samples, are obtained at a 200-Hz sampling rate. The LFPs are produced by linearly combining $N = 2$ independent ERP components $s_1(t)$ and $s_2(t)$, as shown in Fig. 1(a), and an ongoing activity. The single-trial amplitude scaling factors of the two ERP components were drawn independently in each trial from a log-normal distribution with the median 1 and shape parameter 0.5. The single-trial latency shifts of the two ERP components are Gaussian-distributed with zero mean and standard deviations 10 and 20, respectively. The ongoing activity is a second-order AR process with $a_1 = 1.5$, $a_2 = -0.7$, and $\sigma^2 = 1/\text{SNR}$.

Fig. 1(b) shows the AERP when $\text{SNR} = 30$ dB. Inspection of the AERP reveals that there are two main components, and approximately speaking, the first component occurs between 65 and 125 ms, while the second component occurs between 150 and 225 ms. The beginning and the end of each component are marked by vertical dashed lines in the figure. We choose these two AERP segments as the initial estimates of the ERP component waveforms for the dVCA method of [11] and the proposed ASEO method. The dVCA and ASEO estimates of the ERP components are shown in Fig. 1(c) and (d), respectively. As can be seen that the ASEO method provides almost perfect

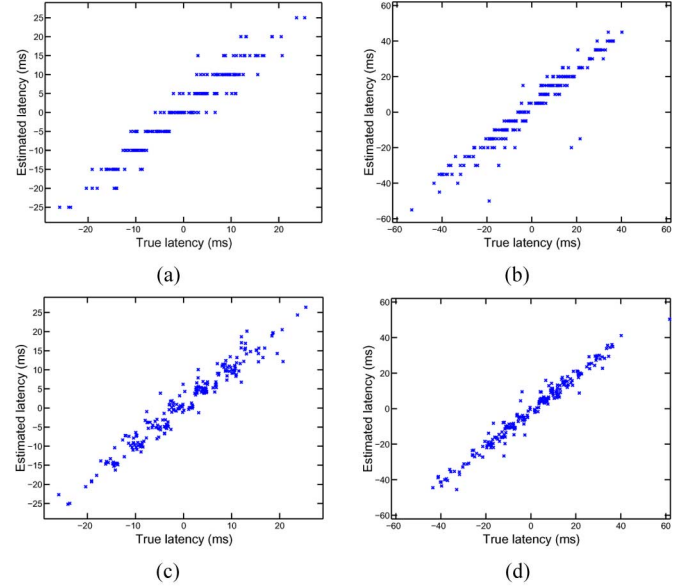


Fig. 2. Comparison of the estimated latencies with their true values for the two ERP component model. (a) dVCA: component 1. (b) dVCA: component 2. (c) ASEO: component 1. (d) ASEO: component 2.

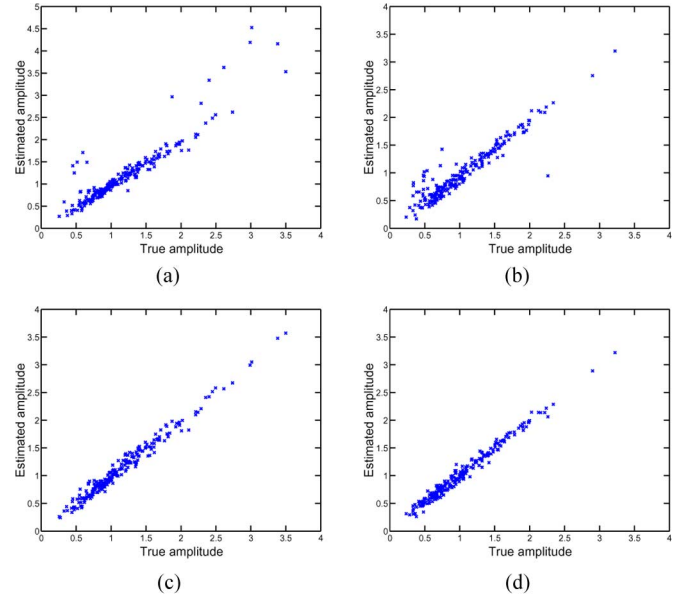


Fig. 3. Comparison of the estimated amplitudes with their true values for the two ERP component example. (a) dVCA: component 1. (b) dVCA: component 2. (c) ASEO: component 1. (d) ASEO: component 2.

replicas of the original ERP waveforms and has better estimation accuracy than dVCA.

Fig. 2(a)–(d) compares the estimated latency shifts, obtained by dVCA and ASEO, respectively, with their true values. As expected, the dVCA estimates of the latency shifts are integer multiples of the sampling period (i.e., 5 ms), which limits the estimation accuracy. Note that the 5-ms limitation does not occur in Fig. 2(c) and (d), which is obtained with the ASEO algorithm. Similarly, as shown in Fig. 3, the ASEO estimates of the amplitude scaling factors are also more accurate than that

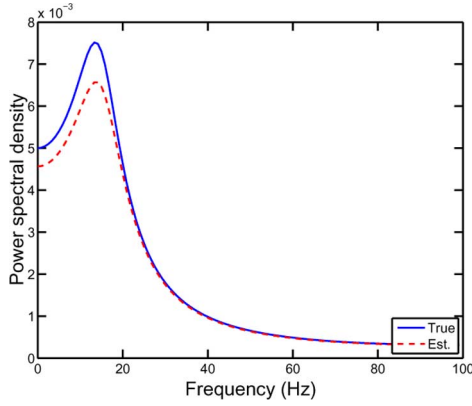


Fig. 4. ASEO PSD estimate of the ongoing activity for the two ERP component example.

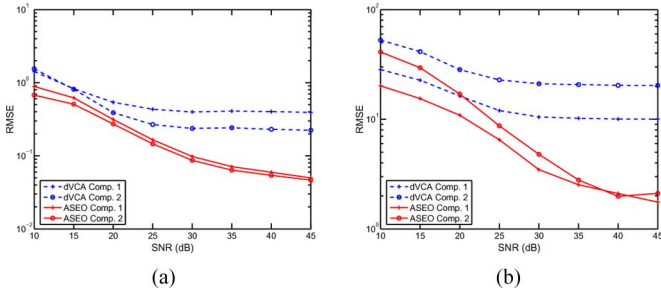


Fig. 5. Empirical root mean square error (RMSE)'s versus SNR for the two ERP component example. (a) Amplitudes. (b) Latencies.

estimated by dVCA. The ASEO estimate of the PSD of the ongoing activity as well as the true PSD are shown in Fig. 4. The ASEO method gives a reasonably good estimate of the PSD of the ongoing activity, even though the estimate is slightly biased downward at low frequencies.

The RMSEs of the estimates of the amplitude scaling factors and the latency shifts as functions of SNR are given in Fig. 5(a) and (b). At low SNR, meaning that the magnitude of the ongoing activity is relatively small compared to the ERPs, the dVCA and ASEO methods give similar performances. However, for $\text{SNR} > 25$ dB, the ASEO method outperforms dVCA significantly with the RMSEs of the former being about ten times smaller than those of the latter for both amplitude scaling factor estimates and latency shift estimates.

2) *Example 2:* The aforementioned example is extended here by considering a more challenging case when there is a third ERP component occurring between 220 and 315 ms. The single-trial amplitude scaling factor and latency shift of the third ERP component were drawn in each trial from a log-normal distribution with the median 1 and shape parameter 0.5 and a Gaussian distribution with the mean 0 and standard deviation 20, respectively. The other simulation parameters in Figs. 1–4 are retained. Figs. 6–8 show the estimates of the ERP waveforms, single-trial amplitude scaling factors, single-trial latency shifts, and the PSD of the ongoing activity by ASEO. Again, ASEO can separate the three ERP components successfully and

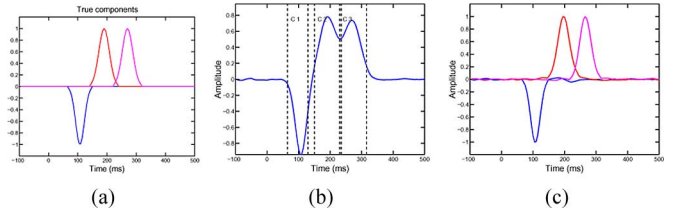


Fig. 6. ERP and AERP waveforms when there are three ERP components. (a) True ERP component waveforms. (b) AERP. (c) ERP component waveforms estimated by ASEO.

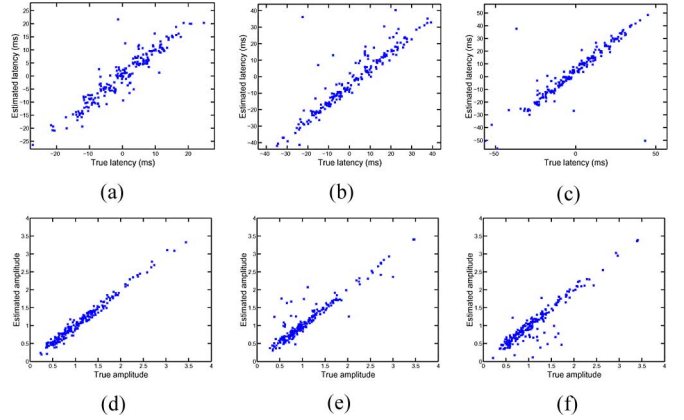


Fig. 7. Comparison of the latencies and amplitudes estimated by ASEO with their true values for the three ERP component example. (a) Latency: component 1. (b) Latency: component 2. (c) Latency: component 3. (d) Amplitude: component 1. (e) Amplitude: component 2. (f) Amplitude: component 3.

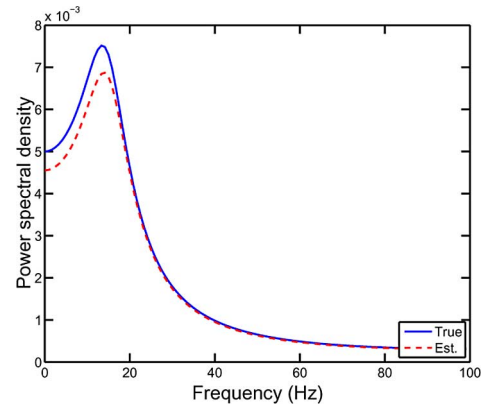


Fig. 8. ASEO PSD estimate of the ongoing activity versus the true value for the three ERP component example.

provide the good estimates of the ERP parameters and PSD of ongoing activity.

3) *Example 3:* In this example, we consider the case where the ERP results from the phase resetting of a portion of the prestimulus ongoing activity. The prestimulus ongoing activity is modeled as the superposition of six AR processes. These six AR processes include: a) five first-order AR processing with AR coefficient a_1 being 0.99, 0.9, 0.4, 0.2, and 0.1, respectively, and the corresponding variance σ^2 of the input white noise being 0.05, 0.1, 0.3, 0.4, and 0.5; and b) a second-order AR process with $a_1 = 1.5$, $a_2 = -0.7$, and $\sigma^2 = 1$. The phase reset

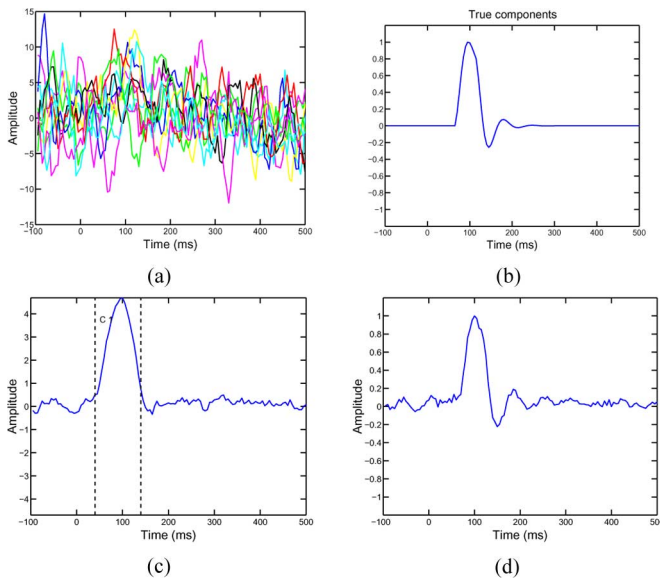


Fig. 9. Single-trial time series, ERP waveform, and AERP waveform when the ERP is generated by the phase reset of a portion of the prestimulus ongoing activity. (a) Superposition of multiple trials of simulated LFPs. (b) Response of the rectangular pulse passing through the second-order AR system. (c) AERP. (d) ERP component waveform estimated by ASEO.

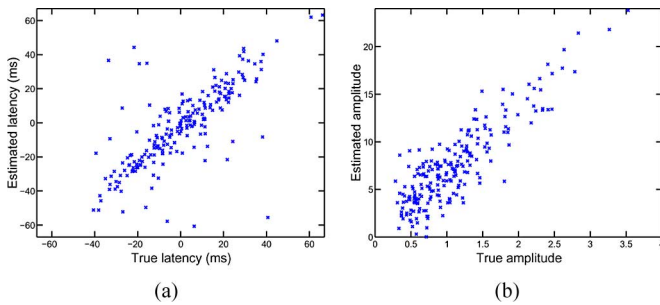


Fig. 10. Comparison of the ERP latencies and amplitudes estimated by ASEO with their true values for the phase reset example. (a) Latency. (b) Amplitude. Note that the amplitude of the input pulse is used here for true amplitude that is proportional to the true ERP amplitude.

of the second-order AR process is accomplished by replacing the noise term by a 50-ms rectangular pulse, whose latency and amplitude vary from trial to trial. The amplitude of the input pulse was drawn in each trial from a log-normal distribution with the median 1 and shape parameter 0.5. The latency shift of the pulse is Gaussian-distributed with mean 65 ms and standard deviation 30 ms, respectively.

Fig. 9(a)–(d) shows the superposition of multiple trials of simulated LFPs, the response of the second-order AR system to the rectangular pulse input that mimics the ERP, AERP obtained by simple averaging, and the estimated ERP by ASEO. Note that from Fig. 9(a), the ERP is not apparent on a single-trial basis due to the presence of significant ongoing activity. Fig. 9(d) shows that ASEO provides a good estimate of the ERP waveform. Fig. 10(a) and (b) compares the estimated ERP latency and amplitude scaling factor with the amplitude and latency of the input rectangular pulse. It is evident that ASEO provides an accurate estimate of the ERP latency and amplitude. Note

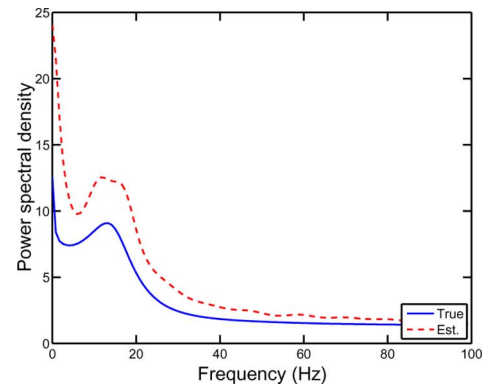


Fig. 11. ASEO PSD estimate of the ongoing activity versus the true value for the phase reset example.

that in this example, the amplitude of the ERP component is not equal to the amplitude of the input pulse. However, they are proportional. Fig. 10(b) demonstrates this proportionality. Fig. 11 shows the estimated PSD of the ongoing activity. The ongoing activity in this example is a summation of several AR processes, which can only be modeled approximately as a single AR process, as is done in the ASEO algorithm. Fig. 11 shows that ASEO can still provide a reasonable estimate of the PSD of the ongoing activity. To further quantify the performance of the PSD estimation, we repeat the same simulation ten times, each time with a different ensemble of random realizations. The peak powers and frequencies of estimated PSD of the ongoing activity are compared with the true values. The peak power is underestimated where the RMSE of the estimated power normalized by the true value is 0.45. The RMSE of the estimated peak frequency is 2.16 Hz, indicating a good recovery of the true peak frequency.

B. Experimental Data

The experimental dataset [39] considered here was originally collected by Dr. R. Nakamura at the Laboratory of Neuropsychology, National Institute of Mental Health. In this experiment, the monkey initiated each trial by depressing and maintaining pressure on a lever, and then indicated his/her perceptual decision of the visual stimulus identity by a GO (lever release) or an NOGO (pressure maintenance) action. LFP data from about 115 ms prior to stimulus onset to 500 ms poststimulus onset were sampled by chronically implanted surface-to-depth bipolar electrodes at multiple cortical sites. The sampling rate was 200 Hz.

The LFPs from two cortical sites for the GO condition were analyzed by the ASEO algorithm. One site is located in the parietal cortex and the other in the primary somatosensory cortex. The AERPs at the two cortical sites are shown in Fig. 12(a) and (b), respectively. Inspection of the AERPs suggests the presence of two main ERP components. We thus chose two nonoverlapping segments of the AERPs, as marked by the dashed vertical lines in Fig. 12(a) and (b), as the initial estimates of the waveforms for the two ERP components. Fig. 12(c) and (d) shows the two estimated ERP components. The early components

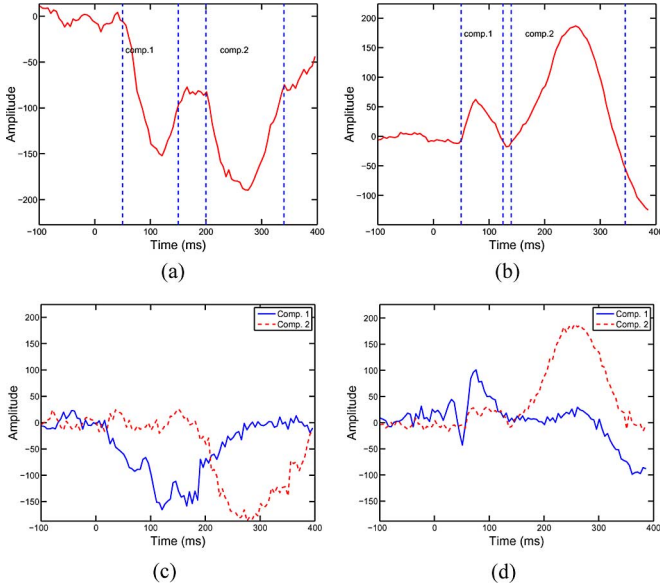


Fig. 12. AERPs and estimated ERP components for experimental data. (a) AERP at the parietal site. (b) AERP at the somatosensory site. (c) Estimated ERP components at the parietal site. (d) Estimated ERP components at the somatosensory site. The vertical axis is in the unit of microvolts.

(component 1) are denoted by solid lines and the late components (component 2) by dashed lines. We chose $K = 20$ as the AR model order for the ongoing activity. The following results were obtained by iterating the ASEO steps five times. In Fig. 12(c) and (d), for comparison purposes, we have amplified the estimated ERP component waveforms to make their peak values equal to the average of the estimated single-trial amplitudes.

Physiologically, there are two distinct stages of neural information processing for the GO trials: sensory activation and motor response. It is reasonable to hypothesize that the early component reflects sensory activation and the late component reflects motor response. Fig. 13(a)–(d) tests this idea by displaying the relationship between the single-trial component latency shift and the corresponding response time (RT). Note that in Fig. 13(b) and (d), significant correlations between RT and the latency shift of the late components are visible. The correlation coefficients are 0.44 and 0.75 with the corresponding p -value less than 0.1%. This indicates that the late component is significantly correlated with the motor phase of the task. On the other hand, no significant RT correlations (with the corresponding correlation coefficients being 0.10 and 0.06 and the corresponding p -values being 10% and 20%, respectively) are observed for the early components in Fig. 13(a) and (c), revealing the sensory nature of the early component.

According to (1), the ongoing activity can be estimated by subtracting out the evoked responses on a trial-by-trial basis from the single-trial LFP data. Fig. 14(a) and (b) shows the ensemble variance of the so-obtained ongoing activities as functions of time (solid curves). For comparison purposes, we also show the ensemble variance of the residual LFPs obtained by simply removing AERPs (dashed curves) from single-trial LFPs. As can be seen, the variance of the residual LFPs obtained

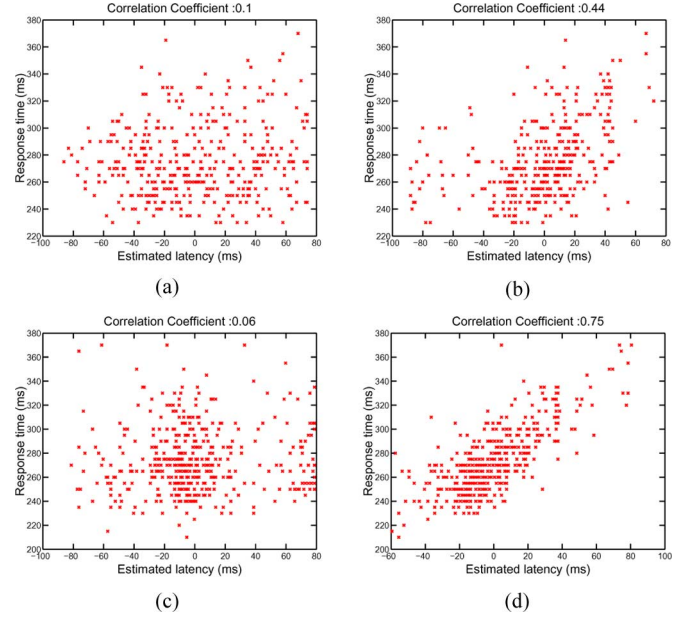


Fig. 13. Correlation between the response time and the single-trial latency shift. The correlation coefficients are given in the parenthesis. (a) Parietal: component 1 (0.10). (b) Parietal: component 2 (0.44). (c) Somatosensory: component 1 (0.06). (d) Somatosensory: component 2 (0.75).

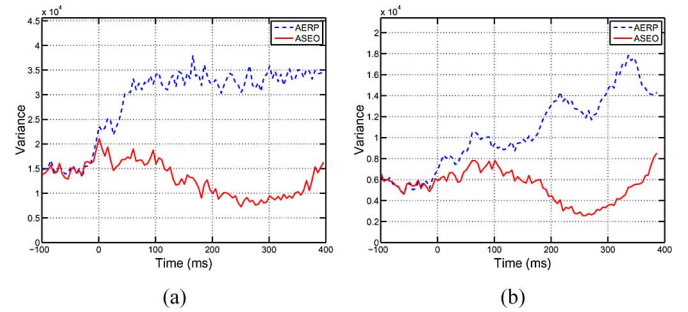


Fig. 14. Variance of the ongoing activity obtained by ASEO (solid) and by simple AERP subtraction (dashed). The normalized RMS of the variance of ongoing activities after subtracting the baseline are 0.29 and 0.30 for ASEO, and 1.14 and 1.22 for AERP. (a) Parietal. (b) Somatosensory. The vertical axis is in the unit of square microvolts.

by simply subtracting the AERPs from single-trial LFPs is increased dramatically in the poststimulus time period. Careful analysis [5] has attributed this large variance to the trial-to-trial variability of the ERPs. This observation predicts that if the ERPs are removed from the LFP signals on a trial-by-trial basis, the variance of the residual ongoing activity should be smaller. This is confirmed by examining the variance of the ongoing activities estimated by ASEO that remain relatively small in the time domain at both recording sites. To quantify this variability, we subtract the baseline (i.e., the variance of the prestimulus signals) from the power of the ongoing activity, compute the RMS of the residual variance over time, and then normalize the obtained RMS by the baseline. The so-obtained normalized RMS represents how well ERPs have been removed from single-trial LFPs. In Fig. 14, the so-obtained normalized RMSs for ASEO are 0.29 and 0.30, respectively, which are much smaller than those obtained by AERP removal (1.14 and 1.22).

VI. CONCLUSION

We have considered the problem of estimating ERPs on a trial-by-trial basis. The LFPs are modeled as a linear combination of multiple ERP components with trial-by-trial varying amplitudes and latencies and an ongoing activity. The ongoing activity is further modeled as an AR random process. Based on this VSPOA model, we then develop an iterative algorithm called ASEO, which iteratively estimates the ERPs in the frequency domain and the ongoing activities in the time domain. The effectiveness of the method is illustrated both for simulated data and experimental data. The simulation examples demonstrate that accurate estimates of the single-trial ERP parameters as well as the PSD of the ongoing activity can be obtained by means of the ASEO method in a variety of realistic situations. Application of the method to experimental data recorded from a monkey performing a visuomotor task reveals two ERP components that correspond to two distinct stages of information processing in the brain: sensory activation (early component) and motor response (late component).

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Luzhou Xu (S'05–M'07) received the B.Eng. and M.S. degrees in electrical engineering from Zhejiang University, Hangzhou, China, in 1996 and 1999, respectively, and the Ph.D. degree in electrical engineering from the University of Florida, Gainesville, in 2006.

From 1999 to 2001, he was with Zhongxing Research and Development Institute, Shanghai. From 2001 to 2003, he was with Philips Research. He is currently with ArrayComm LLC, San Jose, CA. His current research interests include statistical and array

signal processing and their applications.

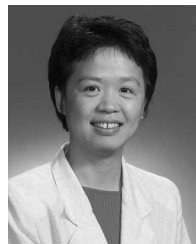


Petre Stoica (SM'91–F'94) received the D.Sc. degree in automatic control from the Polytechnic Institute of Bucharest (BPI), Bucharest, Romania, in 1979, and the Honorary Doctorate degree in science from Uppsala University (UU), Uppsala, Sweden, in 1993.

He is currently a Professor of systems modeling in the Division of Systems and Control, Department of Information Technology, UU. He was a Professor of system identification and signal processing in the Faculty of Automatic Control and Computers, BPI.

He holds visiting positions at Eindhoven University of Technology, Eindhoven, The Netherlands; Chalmers University of Technology, Gothenburg, Sweden (where he held a Jubilee Visiting Professorship); UU; University of Florida, Gainesville; and Stanford University, Stanford, CA. His current research interests include the areas of system identification, time-series analysis and prediction, statistical signal and array processing, spectral analysis, wireless communications, and radar signal processing. He has authored or coauthored nine books, ten book chapters, and some 500 papers in archival journals and conference records. The most recent book he has coauthored, with R. Moses, is *Spectral Analysis of Signals* (Prentice-Hall, 2005). He is a member of the Editorial Boards of six journals: *Journal of Forecasting*, *Signal Processing*, *Circuits, Signals, and Signal Processing*, *Digital Signal Processing Review Journal*, *Signal Processing Magazine*, and *Multidimensional Systems and Signal Processing*. He was a Co-Guest Editor for several special issues on system identification, signal processing, spectral analysis, and radar for some of the aforementioned journals, as well as for the *Institution of Electrical Engineers (IEE) Proceedings*.

Prof. Stoica was the corecipient of the IEEE Advanced Solid-State Photonics (ASSP) Senior Award for a paper on statistical aspects of array signal processing. He was also the recipient of the Technical Achievement Award of the IEEE Signal Processing Society. In 1998, he was the recipient of the Senior Individual Grant Award of the Swedish Foundation for Strategic Research. He was also the corecipient of the 1998 European Association for Signal Processing (EURASIP) Best Paper Award for Signal Processing for a work on parameter estimation of exponential signals with time-varying amplitude, the 1999 IEEE Signal Processing Society Best Paper Award for a paper on parameter and rank estimation of reduced-rank regression, the 2000 IEEE Third Millennium Medal, and the 2000 W. R. G. Baker Prize Paper Award for a paper on maximum-likelihood methods for radar. He was a member of the International Program Committees of many topical conferences. From 1981 to 1986, he was the Director of the International Time-Series Analysis and Forecasting Society. He was a member of the International Federation of Automatic Control (IFAC) Technical Committee on Modeling, Identification, and Signal Processing, a member of the Royal Swedish Academy of Engineering Sciences, an Honorary Member of the Romanian Academy, and a Fellow of the Royal Statistical Society.



Jian Li (S'88–M'90–SM'97–F'05) received the M.Sc. and Ph.D. degrees in electrical engineering from Ohio State University, Columbus, in 1987 and 1991, respectively.

From July 1991 to June 1993, she was an Assistant Professor in the Department of Electrical Engineering, University of Kentucky, Lexington. Since August 1993, she has been with the Department of Electrical and Computer Engineering, University of Florida, Gainesville, where she is currently a Professor. Her current research interests include spectral estimation,

statistical and array signal processing, and their applications.

Prof. Li is a Fellow of the Institution of Electrical Engineers (IEE). She received the 1994 National Science Foundation Young Investigator Award and the 1996 Office of Naval Research Young Investigator Award. Since 2005, she has been a member of the Editorial Board of *Signal Processing* and a Publication of the European Association for Signal Processing (EURASIP). She is currently a member of two of the IEEE Signal Processing Society Technical Committees: the Signal Processing Theory and Methods (SPTM) Technical Committee and the Sensor Array and Multichannel (SAM) Technical Committee.



Steven L. Bressler received the Graduate degree in behavioral neurophysiology and neuroanatomy from the University of California (UC), Berkeley, and the Postdoctoral degree in cognitive science and multivariate time-series analysis from the Electroencephalography (EEG) Systems Laboratory, San Francisco.

He is currently a Cognitive Neuroscientist in the Cognitive Neurodynamics Laboratory, Center for Complex Systems and Brain Sciences, Florida Atlantic University, Boca Raton, where he is engaged

in the study of dynamics of neural information processing during goal-directed behaviors. His current research interests include the development of novel approaches to the study of synchronized oscillations and stimulus-evoked components in cognition.



Xianzhi Shao received the B.S. degree in physics and the M.S. degree in electronics and computers from Shandong University, Jinan, China, and the Ph.D. degree in biomedical engineering from Virginia Commonwealth University, Richmond, in 2005.

In 2005, he was a Postdoctoral Research Associate in the Department of Biomedical Engineering, University of Florida. He is currently a Medical Physicist with OnCure Medical Corporation, Newport Beach, CA. His current research interests include applications of MRI, analysis of event-related potential for

the understanding of cognition, and image-guided radiation therapy.



Mingzhou Ding (M'04) received the B.S. degree in astrophysics from Peking University, Beijing, China, in 1982, and the Ph.D. degree in physics from the University of Maryland, College Park, in 1990.

He is currently a Professor of biomedical engineering at the University of Florida, Gainesville. His current research interests include dynamical systems theory and stochastic processes. He is also involved in cognitive neuroscience and related signal processing problems. In particular, he has pioneered the application of Granger causality to the study of brain

networks using multichannel neurophysiological data. He has authored or coauthored close to 100 journal papers and these papers have received more than 2000 Science Citation Index (SCI) citations.